🧵 Bash String Commands & Operations – Complete Cheat Sheet

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1. STRING VARIABLES

```
str="Hello World"
echo "$str"
```

Always use quotes to preserve spaces.

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2. STRING LENGTH

```
echo ${#str}
```

Example:

```
str="Linux"
echo ${#str}   # 5
```

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3. SUBSTRING (VERY IMPORTANT)

```
${string:position:length}
```

Examples:

```
str="HelloWorld"
echo ${str:0:5}   # Hello
echo ${str:5}     # World
```

---

4. UPPERCASE & LOWERCASE

```
${str^^}    # uppercase
${str,,}    # lowercase
```

Example:

```
str="BaSh"
echo ${str^^}   # BASH
echo ${str,,}   # bash
```

---

5. STRING COMPARISON

```
if [ "$a" = "$b" ]; then
    echo "Equal"
fi
```

Operators:

```
=   !=   -z (empty)   -n (not empty)
```

Examples:

```
[ -z "$str" ]   # empty?
[ -n "$str" ]   # not empty?
```

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6. STRING CONCATENATION

```
a="Hello"
b="World"
c="$a $b"
echo "$c"
```

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7. REMOVE PART OF STRING

Remove prefix

```
${str#pattern}
${str##pattern}
```

Remove suffix

```
${str%pattern}
${str%%pattern}
```

Example:

```
file="report.txt"
echo ${file%.txt}   # report
```

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8. STRING REPLACEMENT

```
${str/old/new}      # first match
${str//old/new}     # all matches
```

Example:

```
str="apple apple"
echo ${str/apple/orange}    # orange apple
echo ${str//apple/orange}   # orange orange
```

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9. CHECK IF STRING CONTAINS SUBSTRING

```
if [[ "$str" == *"word"* ]]; then
    echo "Found"
fi
```

Example:

```
str="Hello Linux"
[[ "$str" == *Linux* ]] && echo "Yes"
```

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10. SPLITTING STRING (IFS)

```
IFS=',' read -ra arr <<< "a,b,c"
echo ${arr[0]}   # a
echo ${arr[1]}   # b
```

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11. READ FULL LINE (INCLUDING SPACES)

```
read -r line
```

From file:

```
while IFS= read -r line
do
    echo "$line"
done < file.txt
```

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12. tr – TRANSLATE / DELETE CHARACTERS

Uppercase:

```
echo "hello" | tr 'a-z' 'A-Z'
```

Remove digits:

```
echo "hello123" | tr -d '0-9'
```

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13. cut – STRING EXTRACTION

```
cut -c3 file.txt
cut -c1-5 file.txt
cut -d':' -f1 /etc/passwd
```

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14. awk – STRING + FIELDS

```
awk '{print $1, $3}' file.txt
```

Uppercase:

```
awk '{print toupper($0)}' file.txt
```

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15. sed – STRING MODIFICATION

```
sed 's/old/new/' file.txt
sed 's/old/new/g' file.txt
```

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16. TRIM SPACES

```
echo "   hi   " | xargs
```

or

```
str="   hi   "
echo "$str" | sed 's/^ *//;s/ *$//'
```

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17. REVERSE STRING

```
echo "hello" | rev
```

18. CHECK STRING IS NUMBER

```
[[ "$str" =~ ^[0-9]+$ ]]
```

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19. STRING IN LOOPS

```
for word in $str
do
    echo "$word"
done
```

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20. COMMON EXAM ONE-LINERS

Print 3rd character from each line:

```
cut -c3 file.txt
```

Convert input to uppercase:

```
read str
echo ${str^^}
```

Replace spaces with commas:

```
echo "$str" | tr ' ' ','
```

EXAM TIPS

- Always use double quotes for strings
- Prefer [[...]] over [...]
- Use read -r for safe input
- Use IFS= when reading lines